Forum on the Integration of Observational and Randomized (FIORD) Workshop 2024

Targeted Learning for Real-World Evidence in Radiation Oncology

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Collaborators

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Vice Chair of ML Department
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Chief Medical Officer, New York Proton Center
President, Proton Collaborative Group

Outline

- **Motivation for RWE Studies in Radiation Therapy**
  Protons (Scattered, Pencil Beam), Photons — Toxicity & Survival
- A General Framework for RWE Generation & Evaluation
- Results & Sensitivity Analysis
- Concluding Remarks

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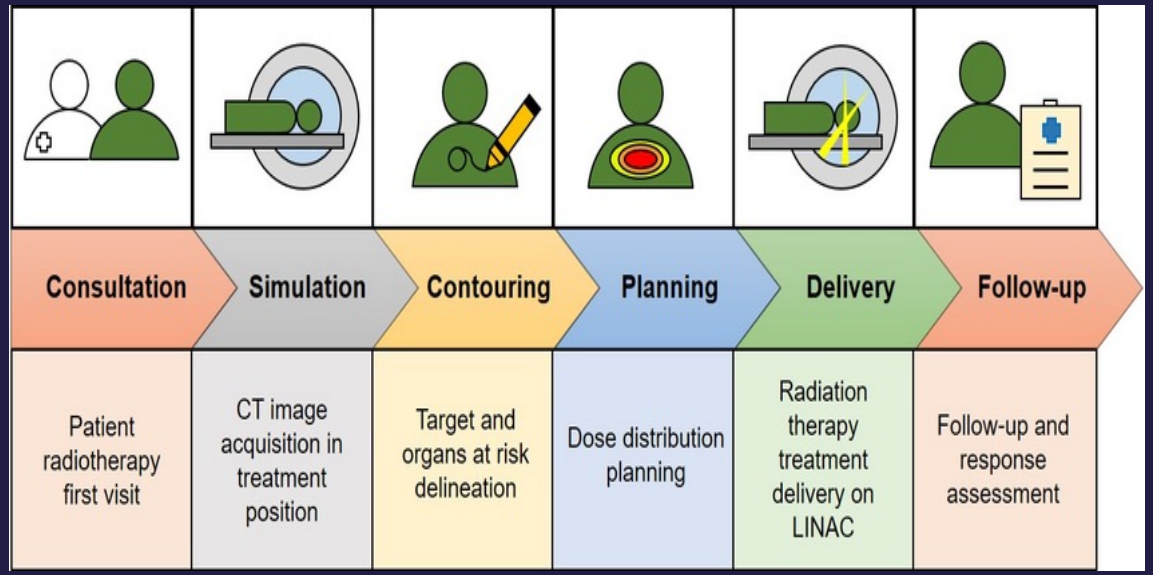
Radiotherapy (RT) is a cornerstone of cancer treatment

>50% of all cancer patients receive RT

Most quantitative oncological field

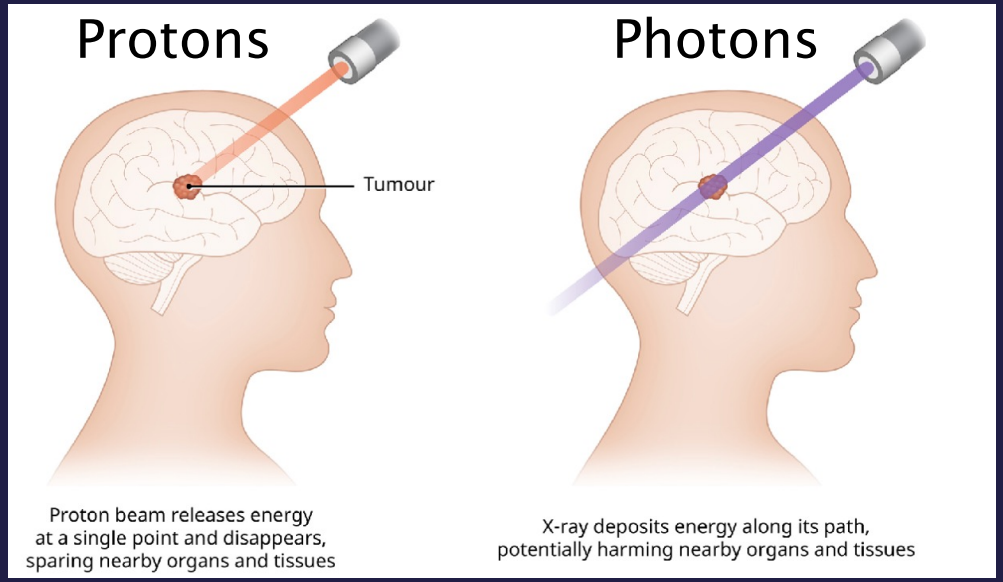
Range of modalities



Marvaso et al. "Virtual and augmented reality as a novel opportunity to unleash the power of radiotherapy in the digital era: a scoping review." *Applied Sciences* (2022)

Range of RT modalities with differing dosimetric profiles and cost

For instance, ***proton pencil beam scanning (PBS)*** offers superior dose precision—sparing organs at risk while delivering the therapeutic dose to the tumor—and is significantly more expensive than both ***photons/X-rays*** and ***proton uniform/passive scattering technology (ST)***



Proton Therapy for Brain and Spine Cancers. https://singaporebrain.org

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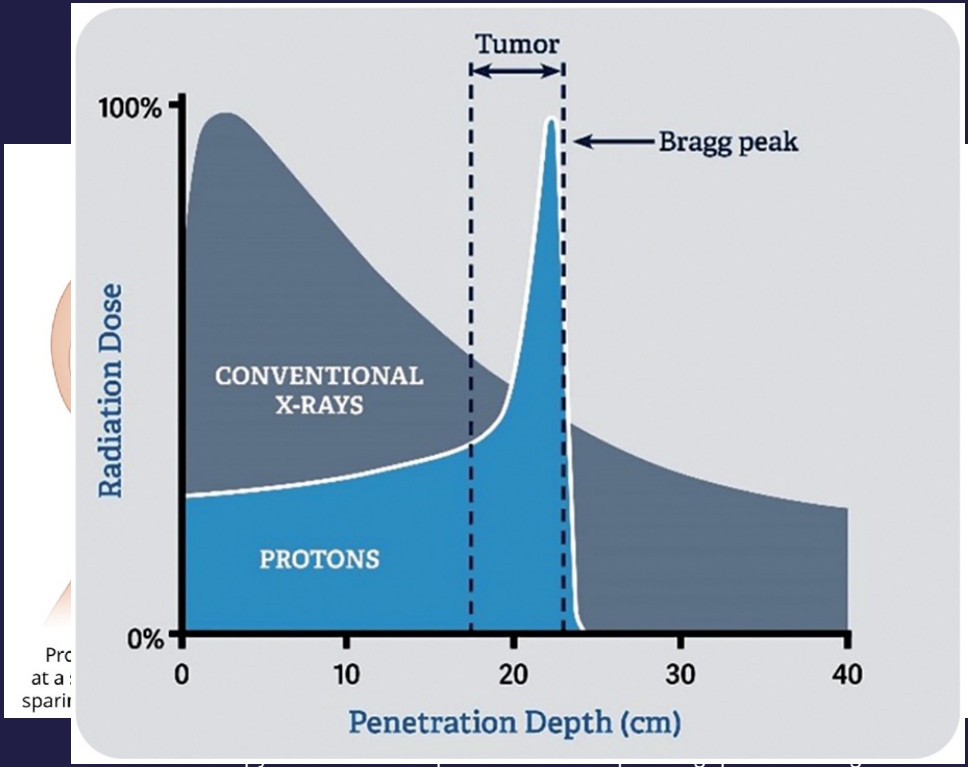

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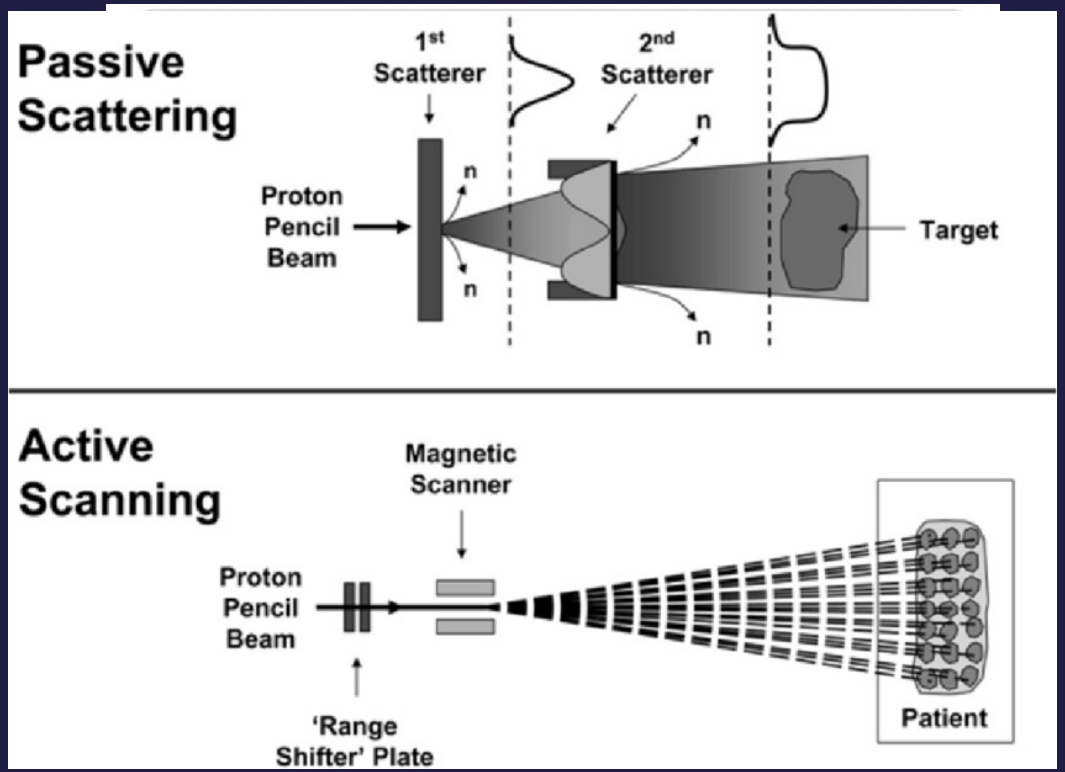

Precision medicine in RT remains underdeveloped

Which patients should receive which RT modality? How to tailor individualized dose prescriptions to manage toxicity and tumor control?

Lacking for many cancer sites like lung cancer, leading cause of cancer deaths in the world

RCTs to establish such evidence cannot keep up with advances in RT technology nor assess causal impacts of toxicity

Ongoing Registry Study for RT Outcomes (NCT01255748)

Proton Collaborative Group (PCG) collects RT data nationwide to better factors that can lead to reduced toxicity and better tumor control

>23,000 patients accrued to date

27 institutions (out of 43 national) offering proton RT

Rich information, including > 137 clinical and dosimetric features from various disease sites

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- **Motivating RWE Study in RT**

Assessing Causal Effects of Proton Radiation & Toxicity on Survival in Stage III Lung Cancer

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Building our DAG

What is the causal effect of PRT on 12-month pneumonitis (≥2)?

Covariates (smoking, race, sex, BMI, age, disease history and prior therapies, cancer stage, tumor size and grade, time from diagnosis to RT, prescription info, etc.)

Toxicity
Pneumonitis (≥2)

Proton Radiation Technology (PRT)
Pencil (PBS) vs. scattered (ST)

time ($t = 0$)

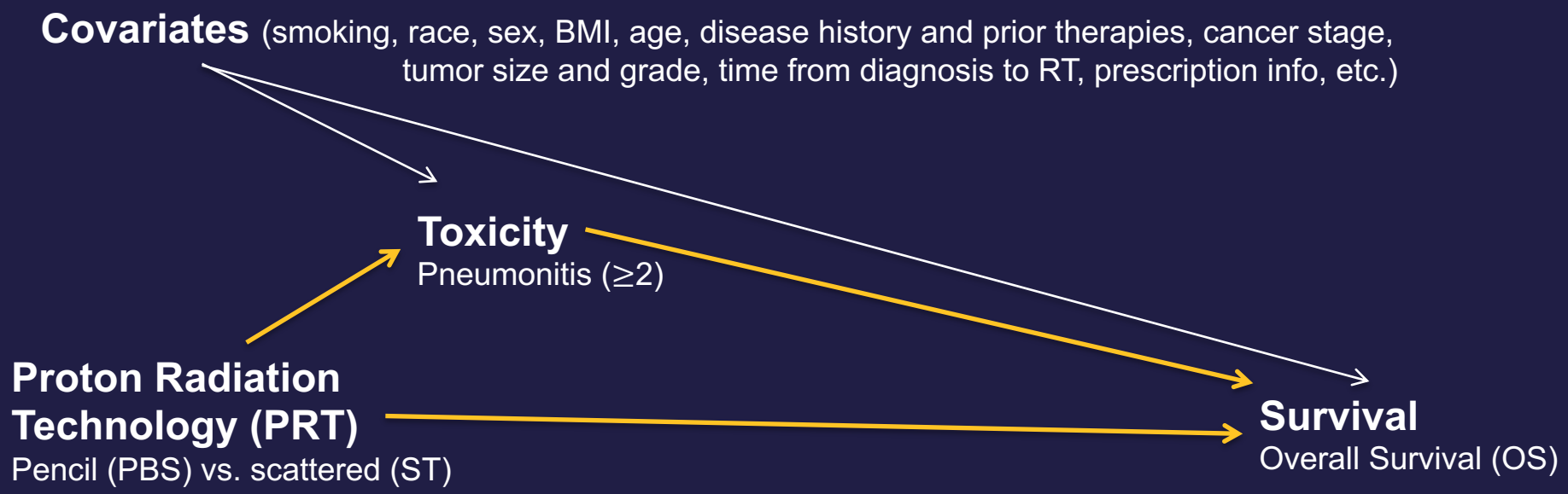
Defining a Causal Estimand

What is the causal risk ratio of PRT on 12-month pneumonitis (≥2)?

$$\psi^{causal} = \frac{P(Tox_{PBT} = 1)}{P(Tox_{ST} = 1)}$$

Building our DAG

Assessing Causal Effects of Proton Radiation & Toxicity on Survival



time (*t = 0*)

PCG Trial Lung Database

965 patients enrolled & treated with protons for lung cancer from 8 institutions, with roughly 25% developing grade ≥2 toxicity (pneumonitis or dyspnea)

39 Total Demographic Features	
Institution	12 Categories
Race	6 Categories
Ethnicity	3 Categories
Weight (kg)	75 (31 – 190)
Height (cm)	170 (65 – 215)
Days from diagnosis to RT	105 (3 – 8148)
Smoking-years	40 (1 – 50)
45 Total Clinical Features	
Treatment Interruption?	Yes (30%), No (70%)
Concurrent Chemo?	Yes (44%), No (56%)
Was it metastatic?	Yes (7.6%), No (92.4%1)
Primary tumor size (cm)	3.5 (0.4 – 9.8)
Grade	5 Categories
Clinical stage	Multi-categories for T,N,M
Pathological stage	Multi-categories for T,N,M
ECOG performance	5 Categories
53 Total Dosimetry Features	
Total proton dose (Gy)	60 (1.2 – 95.0)
Fraction numbers	23 (1 – 57)
Dose metrics for OARS: Esophagus, Heart, Lungs (contralat and ipsalat)	8 Numerical features/OAR: Dmin, Dmax, Davg, D98%, D95%, D50%, d1cc, d2cc

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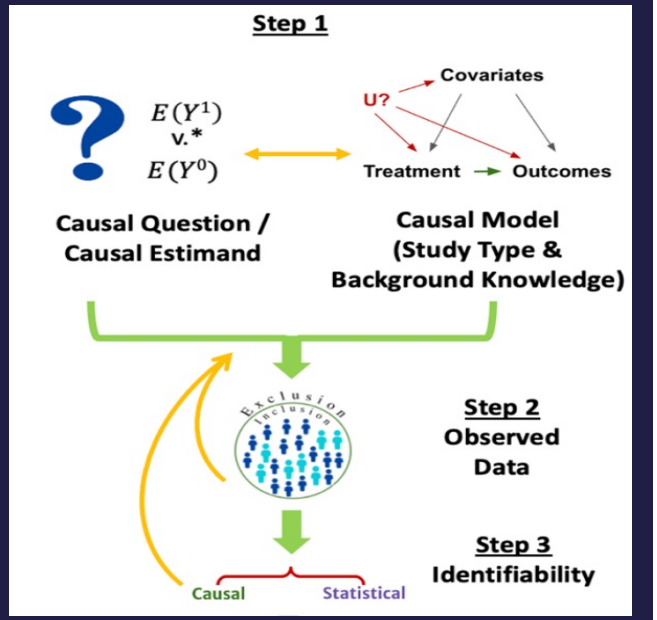
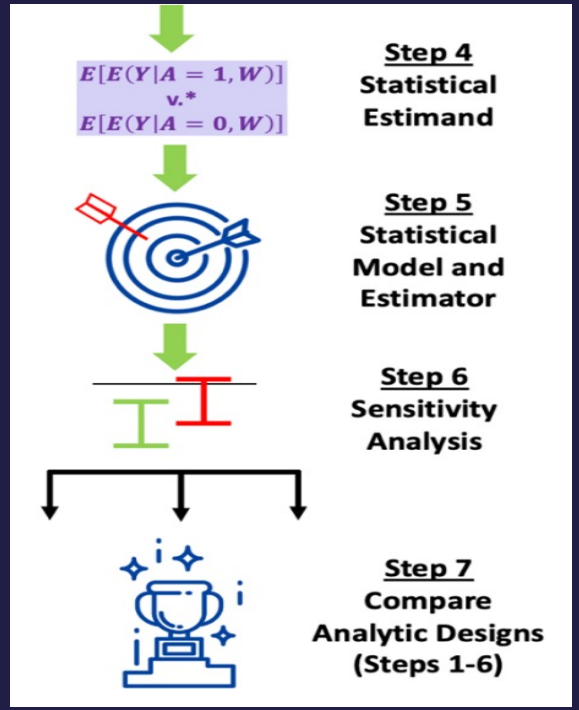
Targeted Learning Roadmap for Causal Inference





Consensus for advancing causal and statistical best practices established by

1. Regulators (FDA, EMA)
2. Clinicians
3. Academic and industry statisticians

Dang et al. "A causal roadmap for generating high-quality real-world evidence" *Journal of Clinical and Translational Science* (2023).

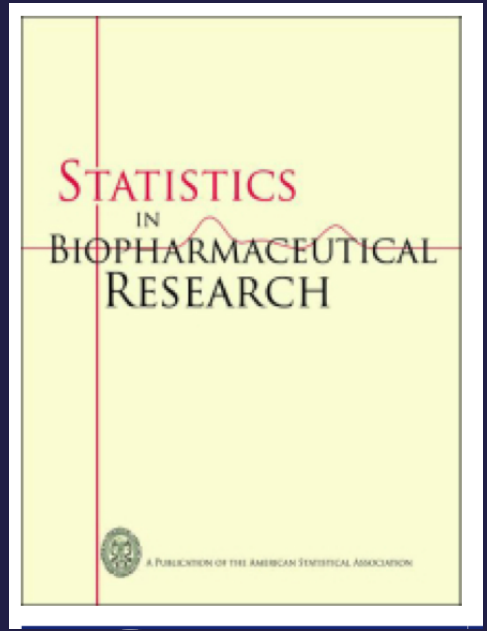
Gruber *et al.*
BMC Medical Research Methodology (2023) 23:178
https://doi.org/10.1186/s12874-023-01998-2

BMC Medical Research Methodology

RESEARCH ARTICLE Open Access

Evaluating and improving real-world evidence with Targeted Learning

Susan Gruber[1*], Rachael V. Phillips[2], Hana Lee[3], John Concato[4] and Mark van der Laan[2]

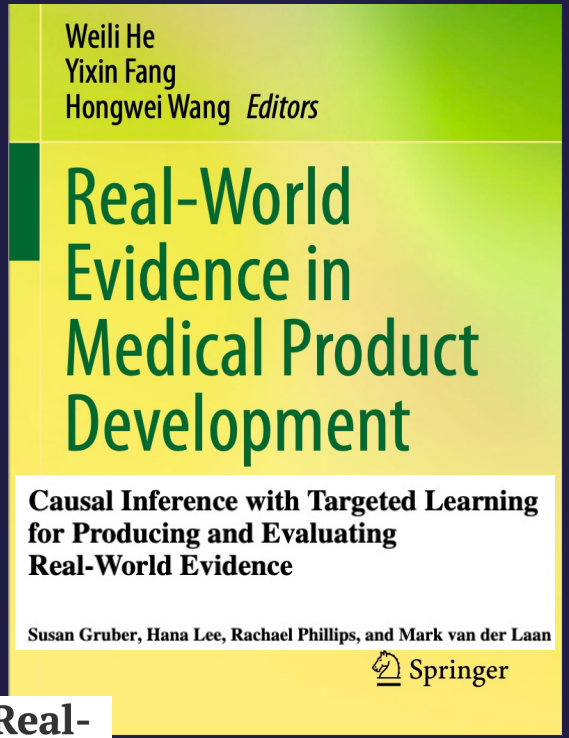


Developing a Targeted Learning-Based Statistical Analysis Plan
Susan Gruber, Hana Lee, Rachael Phillips, Martin Ho & Mark van der Laan

Examples of Applying RWE Causal-Inference Roadmap to Clinical Studies
Martin Ho, Susan Gruber, Yixin Fang, Douglas E. Faris, Pallavi Mishra-Kalyani, David Benkeser &

The Current Landscape in Biostatistics of Real-World Data and Evidence: Causal Inference Frameworks for Study Design and Analysis
Martin Ho, Mark van der Laan, Hana Lee, Jie Chen, Kwan Lee, Yixin Fang, ...show all

Targeted Learning: Toward a Future Informed by Real-World Evidence
Susan Gruber, Rachael V. Phillips, Hana Lee, Martin Ho, John Concato & Mark



3-year FDA-funded Demonstration Project on Targeted Learning

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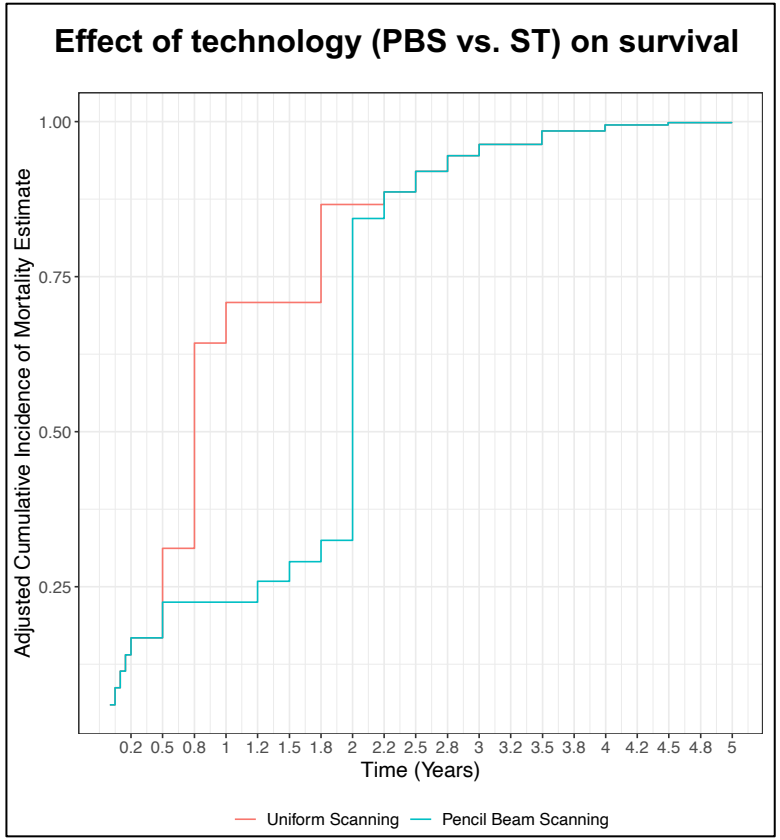
Estimation of Causal Effects of Proton Technology & Toxicity on Survival



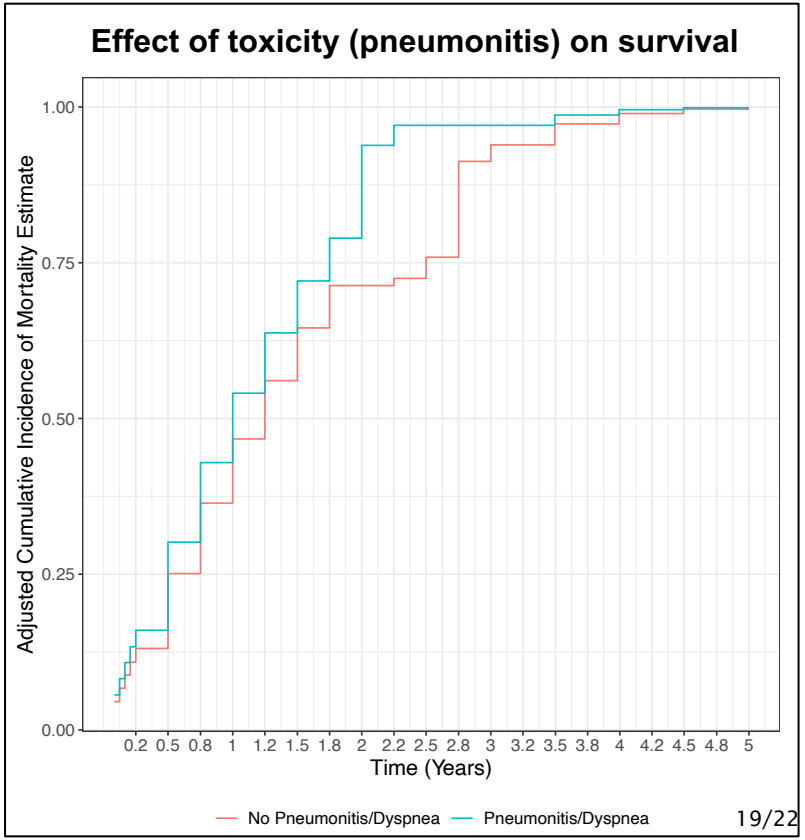
Effect of technology (PBS vs. ST) on toxicity (pneumonitis)

<u>Adjusted Causal Relative Risk</u>

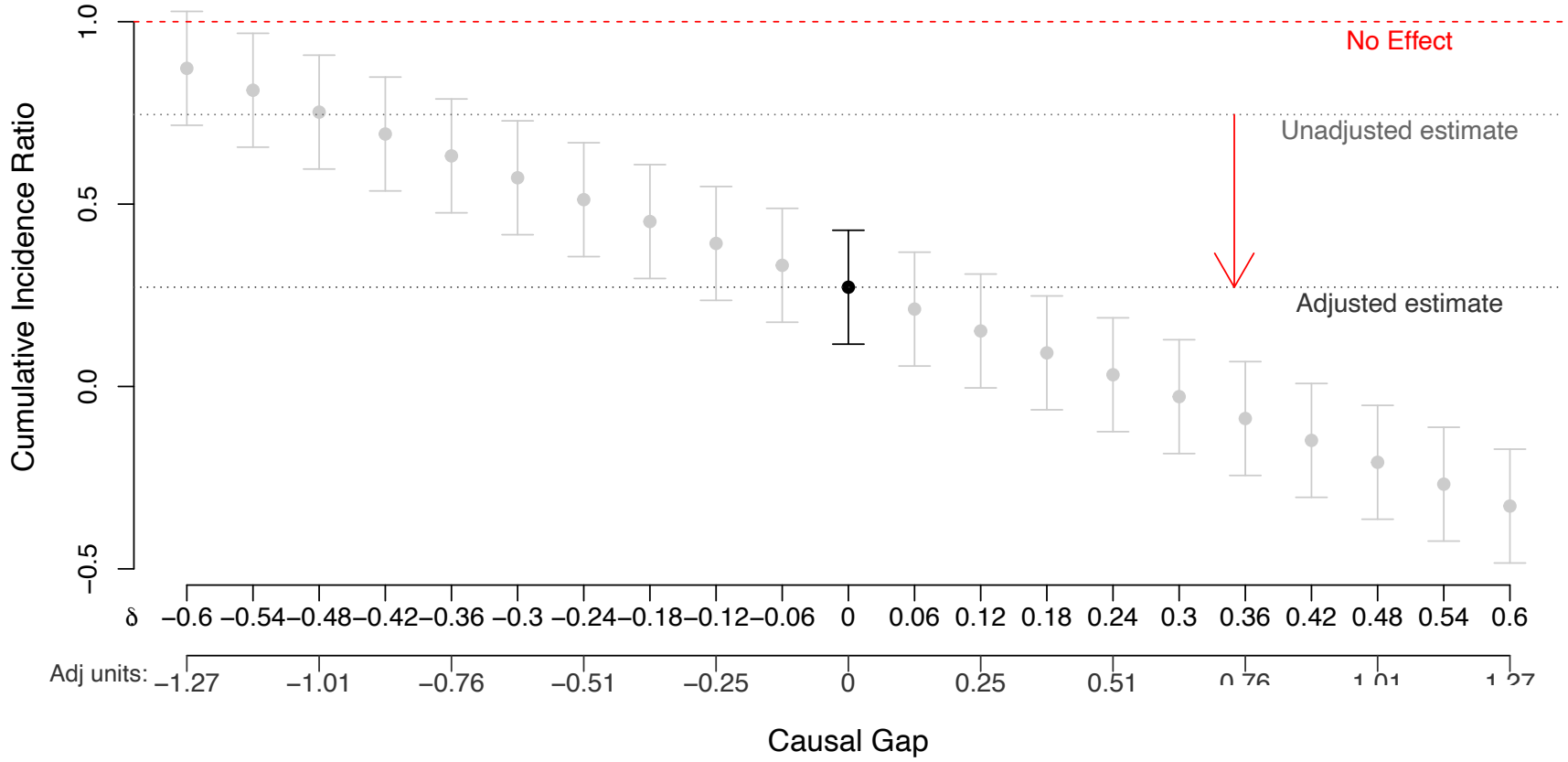
Estimate: 0.48

95% confidence interval: (0.35,0.69)

p-value: 0.0000181

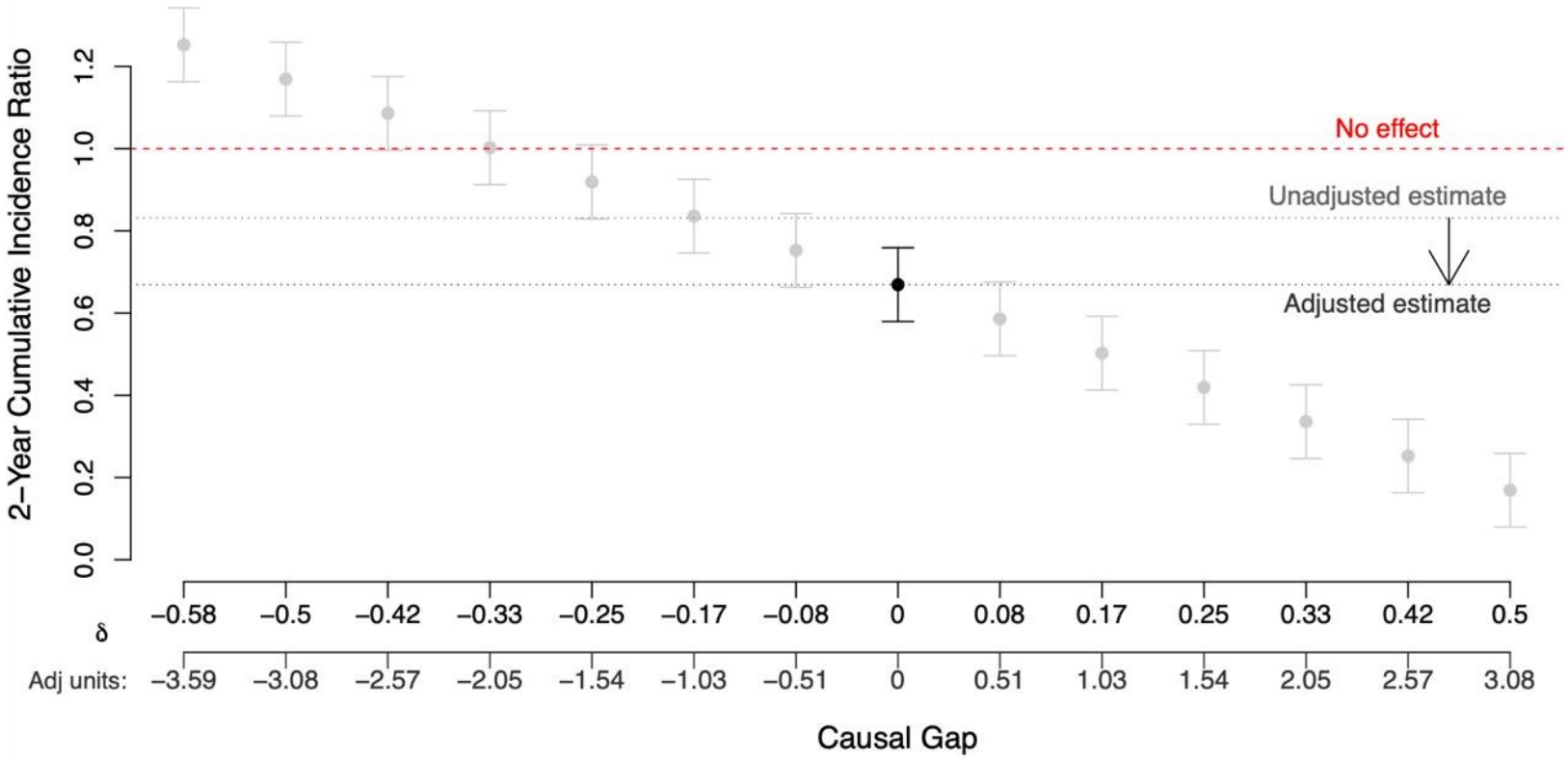


Sensitivity analysis for impact of technology on 1-year survival



It will take hidden confounders — if any — to be 1.27 times as strong as all measured confounders in opposite direction to nullify results of proton technology on 1-year CIR of mortality

Sensitivity analysis for impact of technology on 2-year survival



It will take hidden confounders — if any — to be 1.5 times as strong as all measured confounders in opposite direction to nullify results of proton technology on 2-year CIR of mortality

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Conclusions

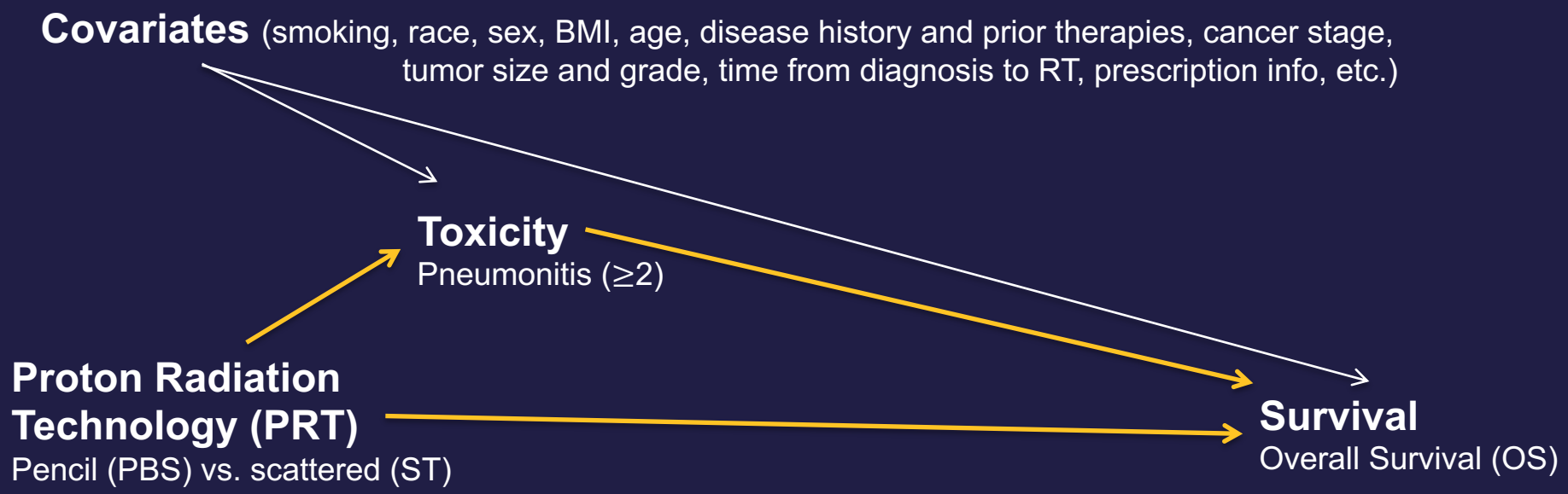
- Proton pencil beam scanning (PBS) leads to better survival as compared to proton scattered technology
- Toxicities lead to worse survival
- Sensitivity analysis show it will take hidden confounders to be much stronger in opposite direction than observed to nullify results of proton technology on 1-year survival
- Targeted Learning can offer a causal insights in settings where clinical trials are not possible or are hard to complete

Limitations and next steps

- Analysis regarding RT-induced toxicity on survival
  - Sensitivity analyses!
  - Immortal time bias induced in this analysis

Building our DAG

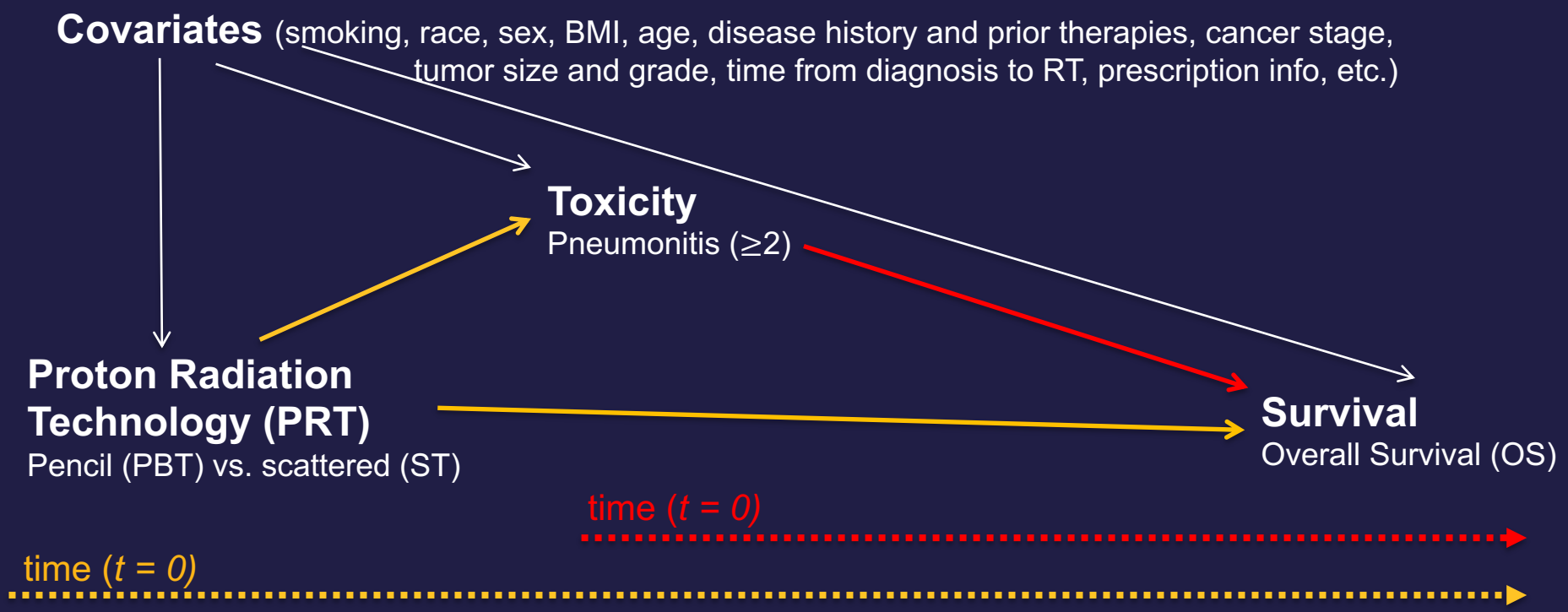
Assessing Causal Effects of Proton Radiation & Toxicity on Survival



time (*t = 0*)

Directed Acyclic Graph (DAG)

Assessing Causal Effects of Proton Radiation & Toxicity on Survival



Limitations and next steps

- Analysis regarding RT-induced toxicity on survival
  - Sensitivity analyses!
  - Immortal time bias induced in this analysis**, it's conservative**
- Causal mediation analysis for the natural direct/indirect effect of proton technology on survival through toxicity
  - Proton technology is an instrumental variable
  - Mediation by RT-induced toxicity is time-varying

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- Analysis regarding RT-induced toxicity on survival
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- Causal mediation analysis for the natural direct/indirect effect of proton technology on survival through toxicity
  - Proton technology is an instrumental variable
  - Mediation by RT-induced toxicity is time-varying
- Toxicity profile as surrogates for survival
  - For example, optimal surrogate based on a score of all toxicity measures so it's highly predictive of survival, entirely blocks the impact of technology on survival

Many future directions

- Sensitivity analyses
- Causal mediation analysis with instrument
- Data collection protocols (e.g. intended treatment plan)
- Data fusion for survival analysis (e.g. RCT+RWD, NRG+PCG)
- Federated learning (e.g. PCG membership)
- Generalizability & transportability (e.g. underrepresented minorities, international)
- Individualized treatment regimes, interpretability